Extended Data Fig. 4 | Cosine similarity on simulation datasets

a, Gene-level cosine similarity between predicted high-dimensional RNA velocity and simulated truth on 42 dyngen datasets. b, Cell-level cosine similarity in PCA2 on the same 42 datasets. The original simulation-comparison layout is retained with current data: alternating grey rows, blue bubble matrices and Median/Mean bars. Dot area is normalized within each panel using the global minimum minus 0.01 and maximum; blue fill shows within-dataset rank (lighter is better). Median/Mean summarize raw dataset scores; their bar widths share a minimum-minus-0.01/max display scale within each panel. Numbers mark the top six ranks, with ties retained. Display normalization does not change the raw cosine values. c, Updated cell-level (upper row) and gene-level (lower row) cosine similarities in the original cellDancer mono-kinetic, transcriptional-boost and multi-forward-branching scenarios at noise 0.2. The original vertical boxplot design is retained; methods are ordered by median within each subplot. Boxes show median and interquartile range, whiskers extend to 1.5 IQR, and outliers are shown. These are the existing common-finite-gene re-evaluation scores provided in the accompanying source workbook, not values digitized from historical artwork. Each scenario has one simulated dataset with 2,000 source cells; the source workbook records shared gene counts and valid denominators. Cells and genes are within-dataset observations, not independent simulation replicates. Methods without valid results are omitted from the corresponding subplot. Saved preprocessing differs by method; alignment does not establish physical velocity units. TopicVelo values in a-b retain their historical-table provenance limitation and are not independently verified native-transition estimates. The full noise-series results are retained in the source-data summary. No model was rerun and the composite ranking is unchanged.